

**Question:**

An investigator is studying the metabolism of an experimental drug that is known to have first order kinetics. Immediately after administering an intravenous dose of the drug to a patient, the serum concentration is 60 U/L. 3 hours later, the serum concentration of the drug is 30 U/L. 9 hours after administration, the serum concentration of the drug is most likely to be which of the following? Answer Choices: A. 5 U/L B. 7.5 U/L C. 15 U/L D. 0 U/L

**Reasoning + Answer:**

<think>The problem describes a situation where the drug follows first-order kinetics, which means that the rate of elimination is proportional to the concentration of the drug in the body. Initially, the serum concentration is 60 U/L, and after 3 hours, it decreases to 30 U/L. This indicates that the drug is being eliminated at a constant rate. To solve this problem, we can use the concept of half-life, which is a characteristic of first-order kinetics. Since the concentration decreased by half in 3 hours, we can infer that the half-life of the drug is approximately 3 hours. After 9 hours, the concentration would have decreased by two half-lives, which means it would be one-quarter of the initial concentration. Therefore, the serum concentration of the drug 9 hours after administration is most likely to be 15 U/L. </think><answer> 15 U/L </answer>

